

MANASWI CHOWDARY MAKKENA

(623) 281-9387 | manaswimakkena@gmail.com | linkedin.com/in/manaswi01 | github.com/manaswichowdary

EDUCATION

Arizona State University

Bachelor of Science in Computer Science (GPA: 3.81)

December 2026

Tempe, AZ

WORK EXPERIENCE

Software Engineer Intern

January 2026 – May 2026

Endless Moments LLC

Austin, TX

- Built AI resume screening engine with bias flagging and real-time WebSocket PTO approval flow; 11 automated tests.
- Delivered 30+ Next.js pages across HR, PTO, and benefits portals across 5 sprints; resolved Docker networking issues.
- Optimized PostgreSQL schema with 28 indexes, 11 constraints, and 14 integrity tests; built centralized error logging.

Software Developer Intern

August 2025 – January 2026

Neuramod

Denver, CO

- Redesigned 4 mobile workflows in FlutterFlow and Firebase, cutting task completion steps by 40% for 200+ beta users.
- Built 6 Cloud Firestore REST endpoints for real-time health data; led schema design, code reviews, and 3 releases.
- Deployed Firebase Cloud Messaging across 3 notification types, increasing daily logging frequency by 25%.

Undergraduate Teaching Assistant | CSE 360: Intro to Software Engineering

January 2026 – May 2026

Arizona State University

Tempe, AZ

- Mentored 100+ students on OOP, design patterns, Agile, and Java; created exam review materials.
- Supported instruction in SDLC concepts including version control, testing strategies, and system design fundamentals.

PROJECTS

MarketLens | Python, FastAPI, Next.js, PostgreSQL, Azure Cognitive Search, OpenAI, AWS Lambda, Docker

- Built data pipeline ingesting 1,000+ job postings daily via Adzuna API on AWS Lambda and EventBridge; sub-100ms PostgreSQL full-text search across 30,000+ records.
- Designed GPT-4o-mini skill extraction pipeline tracking 500+ skills weekly with Azure Cognitive Search semantic ranking.
- Engineered resume scorer returning 0–100 match score against top 50 postings; 15-case pytest suite mocking all APIs.

ResearchAgent | Python, FastAPI, LangChain, Next.js, OpenAI, AWS S3, Chroma, sentence-transformers

- Built LangChain AgentExecutor with 3 custom tools (Tavily, WebBaseLoader, extraction) researching 10+ sources in under 2 minutes with SSE streaming to Next.js.
- Re-implemented agent in raw OpenAI function calling without LangChain, reducing cold-start latency by 35%.
- Replaced OpenAI embeddings with sentence-transformers (all-MiniLM-L6-v2), cutting RAG retrieval from 800ms to 50ms; PDFs stored in AWS S3 via presigned URLs.

Sphere | Next.js, FastAPI, Supabase, OpenAI, LangChain, WebSockets, AWS Lambda, Ollama

- Built GPT-4o chatbot over FastAPI WebSockets with cross-session memory persisted in Supabase JSONB under 1.2s.
- Designed weighted anti-dependency scoring model via AWS Lambda, executing under 800ms, driving session nudges and weekly behavioral trend reports for users.
- Implemented three-tier crisis detection performing real-time NLP keyword analysis before every LLM call, with 988 Lifeline redirection at severity 3; 15-case pytest suite with OpenAI mocking.

Repo Scanner | Python, OpenAI, GitHub REST API, FastAPI, Rich, PyPI

- Published CLI to PyPI (pip install repo-scanner) generating architecture summaries for any GitHub repo under 15 seconds.
- Implemented ast-based file selection cutting token usage by 60%; Railway FastAPI wrapper serves requests under 5 seconds.
- Delivered 3-format output (Rich, JSON, Markdown) with 5-mode fault tolerance and a 10-test pytest suite.

TECHNICAL SKILLS

Languages & Frontend: Python, JavaScript, TypeScript, SQL, Java, C, C++, React, Next.js, Tailwind CSS, Node.js

Backend & AI: FastAPI, REST APIs, WebSockets, OpenAI API, LangChain, RAG, AI Agents, PyTorch, Ollama

Cloud, DevOps & Testing: AWS (S3, Lambda, EC2, EventBridge), Docker, Vercel, Railway, Git, pytest, Jest

Databases: PostgreSQL, Supabase, Redis, Chroma, Cloud Firestore, Firebase, Azure Cognitive Search

Tools & Platforms: FlutterFlow, Firebase Cloud Messaging, Hugging Face, GitHub Actions, PyPI